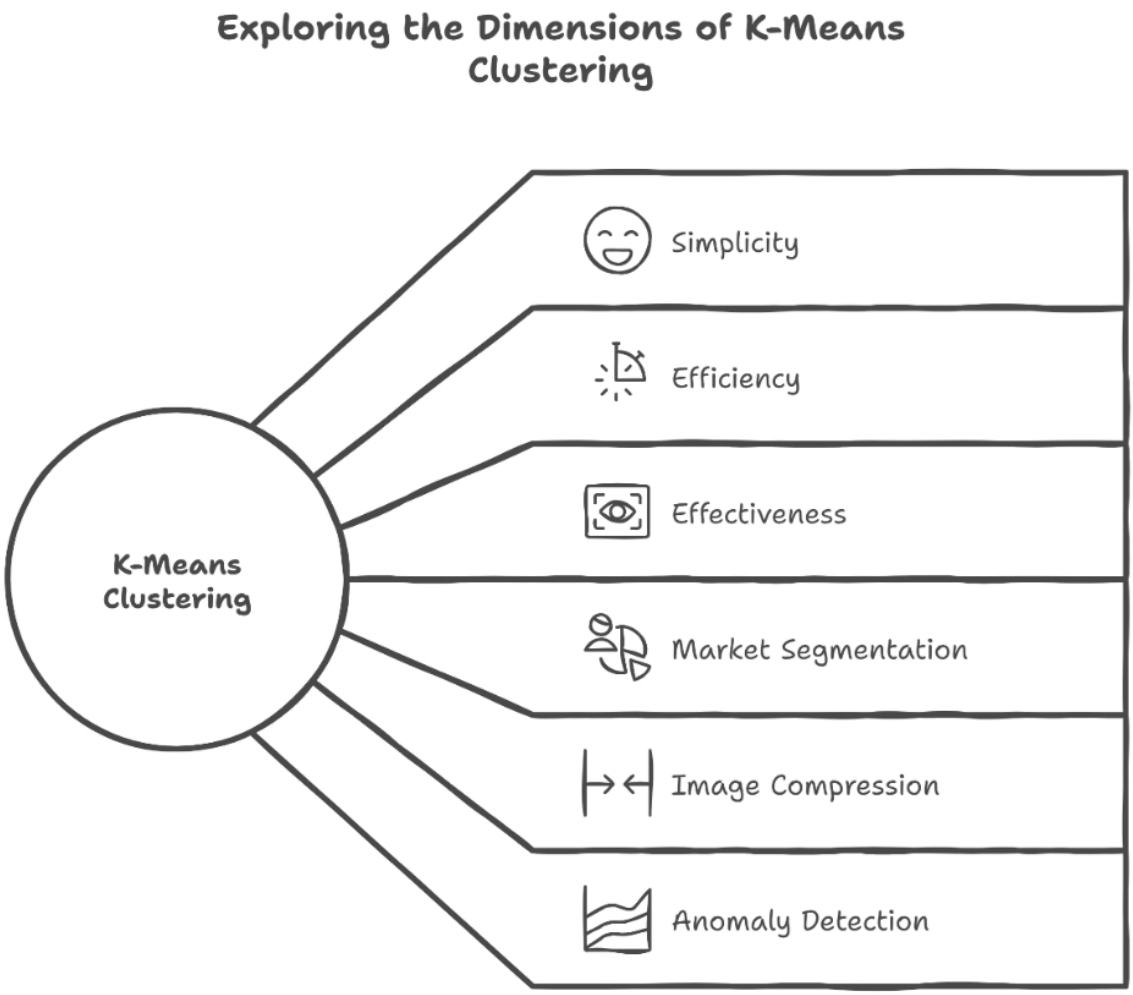


How K-Means Works

K-Means is an iterative algorithm designed to partition a dataset into k distinct, non-overlapping clusters. Each cluster is represented by its centroid, and the algorithm works to minimize the sum of squared distances between data points and their respective cluster centroids. The process can be broken down into the following steps:

1. **Initialization:** The algorithm starts by randomly selecting k initial centroids.
2. **Assignment:** Each data point is assigned to the cluster whose centroid is closest, based on Euclidean distance.
3. **Update:** New centroids are calculated as the mean of all data points assigned to each cluster.